

Discrete Structures Assignment -1

Problem 1:

(a) Truth table of $\neg(p \wedge q) \vee r$ and $(\neg p \vee \neg q) \vee r$:

p	q	r	$\neg(p \wedge q)$	$\neg(p \wedge q) \vee r$	$\neg p \vee \neg q$	$(\neg p \vee \neg q) \vee r$
F	F	F	T	T	T	T
F	F	T	T	T	T	T
F	T	F	T	T	T	T
T	F	F	T	T	T	T
F	T	T	T	T	T	T
T	F	T	T	T	T	T
T	T	F	F	F	F	F
T	T	T	F	T	F	T

Since the truth tables of $\neg(p \wedge q) \vee r$ and $(\neg p \vee \neg q) \vee r$ match, they are equivalent.

(b)

p	q	$p \vee q$	$p \vee \neg q$	$(p \vee q) \wedge (p \vee \neg q)$
F	F	F	T	F
F	T	T	F	F
T	F	T	T	T
T	T	T	T	T

Since truth tables of p and $(p \vee q) \wedge (p \vee \neg q)$ match, they are equivalent.

Problem 2:

- (a) $\forall x (C(x) \rightarrow P(x))$
- (b) $\exists x (P(x) \wedge \neg C(x))$
- (c) $\forall x (P(x) \vee C(x))$
- (d) $\exists x (C(x) \oplus P(x))$

Problem 3:

(a) $\forall n \in \mathbb{Z} ((O(n) \wedge (n > 1)) \rightarrow \exists p \in \mathbb{Z} (P(p) \wedge (p \leq n) \wedge (p|n)))$

(b) Negation of above ~~statement~~ expression is:

$$\begin{aligned} & \neg \forall n \in \mathbb{Z} ((O(n) \wedge (n > 1)) \rightarrow \exists p \in \mathbb{Z} (P(p) \wedge (p \leq n) \wedge (p|n))) \\ \equiv & \exists n \in \mathbb{Z} \neg ((O(n) \wedge (n > 1)) \rightarrow \exists p \in \mathbb{Z} (P(p) \wedge (p \leq n) \wedge (p|n))) \\ \equiv & \exists n \in \mathbb{Z} ((O(n) \wedge (n > 1)) \wedge \neg \exists p \in \mathbb{Z} (P(p) \wedge (p \leq n) \wedge (p|n))) \\ \equiv & \exists n \in \mathbb{Z} ((O(n) \wedge (n > 1)) \wedge \forall p \in \mathbb{Z} \neg (P(p) \wedge (p \leq n) \wedge (p|n))) \\ \equiv & \exists n \in \mathbb{Z} (O(n) \wedge (n > 1) \wedge \forall p \in \mathbb{Z} (\neg P(p) \vee (p > n) \vee (p \nmid n))) \end{aligned}$$

(c) Translation of expression in part (b):

There exists an integer n such that n is odd, and greater than 1, and for every integer p , either p isn't prime, or p is greater than n , or p doesn't divide n .

More naturally,

There exists an odd integer n greater than 1, which has no prime divisors.

~~But~~ But it isn't easily translatable to propositional logic.

(d) The error is that student used "there exists a prime p " instead of "for all integers p such that..." required for correct negation since $\neg \exists p \in N(\dots) \equiv \forall p \in N \neg(\dots)$.

Problem 4:

We need to prove that $(1+x)^n \geq 1+nx \forall n \in \mathbb{N}$ where $x \in \mathbb{R}$
 $x > -1$

We can use induction.

$$P(n): (1+x)^n \geq 1+nx$$

Base case:

$$P(0): (1+x)^0 \geq 1+0$$

$1 \geq 1$

$\therefore P(0)$ is true

Induction hypothesis:

$$P(k): (1+x)^k \geq 1+kx$$

Let, it be true

Inductive step:

$$P(k+1): (1+x)^{k+1} \geq 1 + (k+1)x$$

We need to prove $P(k) \rightarrow P(k+1)$.

$$\text{Now, } (1+x)^k \geq 1 + kx \quad [\text{from induction hypothesis}]$$

Since $1+x > 0$, multiplying both sides by $1+x$ retains the inequality.

$$\begin{aligned} \therefore (1+x)^{k+1} &\geq (1+kx)(1+x) \\ \Rightarrow (1+x)^{k+1} &\geq 1 + kx + x + kx^2 \\ \Rightarrow (1+x)^{k+1} &\geq 1 + (k+1)x + kx^2 \end{aligned}$$

$$\text{We know, } kx^2 \geq 0. \quad [\because k \geq 0 \text{ and } x^2 \geq 0]$$

$$\Rightarrow (1+x)^{k+1} \geq 1 + (k+1)x$$

$$\therefore P(k) \rightarrow P(k+1)$$

$$\therefore (1+x)^n \geq 1 + nx \quad \text{is true } \forall n \in \mathbb{N} \text{ by mathematical induction.}$$

$$x \in \mathbb{R}, x > -1$$

QED

Problem 5:

(a) To prove or disprove: $\forall x \in \mathbb{R}$, if x is rational, then $x^4 + x^2 + 1 > 0$

~~Now~~ Now, $x \in \mathbb{R} \Rightarrow x^2 \geq 0 \Rightarrow x^4 \geq 0$ — (1)

And, $1 > 0$ — (2)

From (1) and (2), $x^4 + x^2 + 1 > 0 \quad \forall x \in \mathbb{R}$.

It is true for both rational and irrational x .

$\therefore$ the statement is proved.

(b) To prove or disprove: $\forall n \in \mathbb{N}$, $n^2 - n + 41$ is prime.

If we take $n = 41$, $n^2 - n + 41 = 41^2 - 41 + 41 = 41^2$, which is divisible by 41.

$\therefore$ it isn't a prime in this case.

$\therefore$ the statement $n^2 - n + 41$ is a prime $\forall n \in \mathbb{N}$ is ~~not~~ shown false by taking a counter-example.

(c) To prove: If p and $p+1$ are both prime and $p > 2$, p^2 is even.

Let, p is a prime and $p > 2$.

$\therefore p$ is an odd number because 2 is the only even prime.

$\therefore p+1 > 2$ and $p+1$ is even.

But every ^{even} number greater than 2 isn't prime.

$\therefore$ the statement " p and $p+1$ are both prime and $p > 2$ " is false by contradiction.

$\therefore$ the statement "if p and $p+1$ are both prime and $p > 2$, then p^2 is even" is always true vacuously.

(d) To prove: $\forall p, q \in \mathbb{Q}, p < q$, there exists $r \in \mathbb{Q}$ st $p < r < q$.

$$x := \frac{q-p}{2} \text{ is a rational [half of difference of numbers]}$$

$$\text{Since } q > p, x > 0. \quad \text{--- ①}$$

$$\text{Now, } p+x = p + \frac{q-p}{2} = \frac{p+q}{2}$$

$$\text{And, } q-x = q - \frac{q-p}{2} = \frac{p+q}{2}$$

$$\begin{array}{ll} \text{And also, } p+x > p & [\text{from ①}] \\ q-x < q & [\text{from ①}] \end{array}$$

$$\text{Then, } r := \frac{p+q}{2} \text{ satisfies } p < r < q \text{ and } r \in \mathbb{Q}.$$

$\therefore$ the statement is shown true by construction $\Rightarrow$ proof.

Problem 6:

We can prove it using the technique of checkerboard coloring.

We color the 8×8 chessboard by basic checkerboard pattern. It will contain 32 squares of black and white each.

For tiling it, equal number of squares of each color must be covered.

For T-tetrominoes:

It always covers 3 squares of one color and 1 of other

$$\begin{aligned}\therefore \text{color imbalance/difference} &= 3-1 \text{ or } 1-3 \\ &= 2 \text{ or } -2 \\ &= \pm 2\end{aligned}$$

For square tetrominoes:

It always covers 2 squares of both colors.

$$\therefore \text{color difference} = 0$$

$$\therefore \text{total color difference, } D = \overbrace{(\pm 2) + (\pm 2) + \dots + (\pm 2)}^{15 \text{ times}} + 0$$

$$\frac{D}{2} = \underbrace{(\pm 1) + (\pm 1) + \dots + (\pm 1)}_{15 \text{ times}}$$

Sum of odd number of ± 1 can never be 0.

$\therefore D$ can never be zero.

$\therefore$ chessboard can never be tiled by 15 T-tetrominoes and 1 square tetrominoes.

Problem 7:

To prove: $\forall n \in \mathbb{N}, n^7 - n$ is divisible by 42.

$$42 = 7 \times 3 \times 2$$

$\therefore$ we need to prove that $n^7 - n$ is divisible by all of 7, 3 and 2.

Divisibility by 2 and 3:

$$\begin{aligned} n^7 - n &= n(n^6 - 1) = n(n^2 - 1)(n^4 + n^2 + 1) \\ &= (n-1)n(n+1)(n^4 + n^2 + 1) \end{aligned}$$

n and $(n+1)$ are consecutive natural numbers. $\therefore$ one of them is even. $\therefore n^7 - n$ is divisible by 2.

$(n-1)$, n and $(n+1)$ are 3 consecutive natural numbers $\therefore$ one of them is a multiple of 3. $\therefore n^7 - n$ is divisible by 3.

Divisibility by 7:

We can use induction.

$$P(n): 7 \mid n^7 - n$$

Base case:

$$P(0): 7 \mid 0$$

$P(1)$ is true.

Induction hypothesis:

$$P(k): 7 \mid k^7 - k$$

Let, it be true.

Inductive step:

We need to prove $P(k) \rightarrow P(k+1)$: $7 \mid (k+1)^7 - (k+1)$

$$\begin{aligned} \text{Now, } (k+1)^7 - (k+1) &= \left[{}^7C_0 + {}^7C_1 k + {}^7C_2 k^2 + {}^7C_3 k^3 \right. \\ &\quad \left. + {}^7C_4 k^4 + {}^7C_5 k^5 + {}^7C_6 k^6 + {}^7C_7 k^7 \right] \\ &\quad - (k+1) \\ &= (1 + 7k + 21k^2 + 35k^3 + 35k^4 + 21k^5 + 7k^6 + k^7) \\ &\quad - (k+1) \\ &= (k^7 - k) + 7(k + 3k^2 + 5k^3 + 5k^4 + 3k^5 + k^6) \\ &= (k^7 - k) + 7k' \end{aligned}$$

Since, $7 \mid (k^7 - k)$ and $7 \mid 7k'$, $7 \mid (k+1)^7 - (k+1)$.
induction hypothesis

$$\therefore P(k) \rightarrow P(k+1)$$

$\therefore 7 \mid n^7 - n \quad \forall n \in \mathbb{N}$ by mathematical induction.

$2 \mid n^7 - n$ and $3 \mid n^7 - n$ and $7 \mid n^7 - n \Rightarrow 42 \mid n^7 - n$.
 $\therefore$ proved.

Problem 8:

Let, x and y are integers.
 $2x^2$ is ~~even~~ even and 14 is even.

$\therefore 5y^2$ must be even.
 $\Rightarrow y^2$ must be even
 $\Rightarrow y$ must be even

$$\therefore y = 2k \quad k \in \mathbb{Z}$$

Then, $2x^2 + 20y^2 = 14$
 $\Rightarrow x^2 + 10y^2 = 7$

If $y = 0$,

$$x^2 = 7$$

$$x = \sqrt{7} \notin \mathbb{N}$$

This is a contradiction

If $y \neq 0$,

$$y^2 \geq 1$$

$$10y^2 \geq 10$$

$$10y^2 > 7$$

$$x^2 + 10y^2 > 7$$

$\therefore$ a solution isn't possible in this case.

Since the ~~only~~ case causes contradiction or isn't possible, our assumption is wrong

$\therefore$ there are no solutions in integers x and y to eqn
 $2x^2 + 5y^2 = 14$.